



ADVANCED PHYSICAL DESIGN SKILLTEST BOOK

Designed for VLSI Engineers Preparing for Core
Physical Design Interviews

- ✓ *Industry Curated Advanced Questions*
- ✓ *Designed for Core PD Roles*
- ✓ *Focused on Timing Closure & Power Integrity*
- ✓ *Explanations with Concept Clarity*



Question 1

During floorplanning, what is the primary purpose of creating a soft blockage?

- A) To completely prevent placement and routing
- B) To prevent only routing in that region
- C) To discourage cell placement while allowing limited placement if needed
- D) To reserve area exclusively for macros

Answer: C

Explanation: A soft blockage discourages standard cell placement but does not strictly prohibit it. Tools may use the region if congestion becomes severe.

Question 2

Which of the following techniques is most effective in reducing IR drop in advanced nodes?

- A) Increasing cell density
- B) Reducing metal width of power rails
- C) Adding more power straps and vias
- D) Increasing clock frequency

Answer: C

Explanation: IR drop is reduced by lowering resistance in the power network. Adding wider straps and more vias reduces effective resistance.

Question 3

In CTS (Clock Tree Synthesis), why is useful skew intentionally introduced?

- A) To increase clock latency
- B) To reduce dynamic power
- C) To improve setup timing on critical paths
- D) To eliminate hold violations completely

Answer: C

Explanation:

Useful skew redistributes clock arrival times to improve setup timing margins on critical paths.

Question 4

What is the major impact of double patterning in 7nm and below technologies?

- A) Reduced routing congestion
- B) Increased timing margins
- C) Additional coloring constraints during routing
- D) Elimination of antenna violations

Answer: C

Explanation:

Double patterning introduces mask coloring constraints that the router must honor to avoid lithography conflicts.

Question 5

Which parameter primarily determines electromigration reliability?

- A) Switching activity
- B) Current density
- C) Clock skew
- D) Cell height

Answer: B

Explanation: Electromigration depends on the current density (J) represented as ($J = I/A$). High current density accelerates metal atom migration.

Question 6

In MCMM (Multi-Corner Multi-Mode) analysis, why are slow corners used for setup checks?

- A) Because delay is minimum
- B) Because delay is maximum
- C) Because leakage is minimum
- D) Because routing is optimized

Answer: B

Explanation:

Slow corners increase cell and interconnect delays, making setup violations more likely.

Question 7

Which of the following best describes the purpose of tap cells?

- A) Improve clock skew
- B) Prevent latch-up by connecting substrate to power rails
- C) Reduce routing congestion
- D) Improve hold timing

Answer: B

Explanation:

Tap cells connect wells/substrate to power/ground to avoid latch-up conditions.

Question 8

During detailed routing, which violation is most likely fixed by increasing metal spacing?

- A) Antenna violation
- B) DRC spacing violation
- C) Hold violation
- D) Setup violation

Answer: B

Explanation:

DRC spacing violations occur when wires are too close. Increasing spacing fixes this.

Question 9

What is the primary reason for inserting filler cells?

- A) Improve timing
- B) Improve routing efficiency
- C) Maintain well continuity and satisfy DRC
- D) Reduce IR drop

Answer: C

Explanation:

Filler cells ensure proper well continuity and meet design rule requirements.

Question 10

Which scenario most likely causes hold violations after CTS?

- A) Large combinational delay
- B) Increased clock uncertainty
- C) Reduced clock skew
- D) Faster clock arrival at capture flop

Answer: D

Explanation:

If capture clock arrives earlier than launch clock, data may violate hold timing.

Question 11

In congestion analysis, which metric is commonly used to estimate routability?

- A) WNS
- B) TNS
- C) GCell overflow
- D) Clock latency

Answer: C

Explanation:

Global routing cells (GCells) measure routing resource demand vs availability.

Question 12

Why are shield wires used in high-speed routing?

- A) Reduce IR drop
- B) Reduce crosstalk noise
- C) Improve setup margin
- D) Fix antenna violations

Answer: B

Explanation:

Shield wires reduce capacitive coupling between aggressor and victim nets.

Question 13

What is the key difference between global routing and detailed routing?

- A) Global routing assigns exact track locations
- B) Detailed routing estimates congestion
- C) Global routing assigns approximate paths
- D) Detailed routing ignores DRC

Answer: C

Explanation:

Global routing provides coarse path planning; detailed routing assigns exact tracks and vias.

Question 14

Which technique is used to fix antenna violations without changing routing topology?

- A) Buffer insertion
- B) Metal hopping
- C) Increasing clock skew
- D) Reducing power straps

Answer: B

Explanation:

Metal hopping moves routing to higher layers to reduce antenna ratio.

Question 15

At advanced nodes (5nm and below), which effect significantly impacts timing closure?

- A) Wire resistance dominance
- B) Gate delay dominance
- C) Temperature independence
- D) Reduced variability

Answer: A

Explanation: At advanced nodes, interconnect resistance dominates delay.

Question 16

During post-route timing analysis, setup violations are seen only in RC worst corner but not in slow-slow corner. What is the most probable reason?

- A) Cell delay dominates timing
- B) Wire RC dominates timing
- C) Clock skew is negative
- D) Hold margin is high

Answer: B

Explanation:

RC worst corner models worst interconnect parasitics. If violations appear only here, wire delay is dominating.

Question 17

In advanced nodes, why is via resistance becoming a major concern?

- A) Via count is reduced
- B) Via dimensions shrink, increasing resistance
- C) Metal layers increase
- D) Routing congestion reduces

Answer: B

Explanation: As via dimensions shrink, resistance increases, impacting timing and IR drop.

Question 18

If hold violations increase significantly after routing, what is the most likely reason?

- A) Decrease in setup slack
- B) Extraction of parasitics
- C) Increase in clock period
- D) Reduction in clock latency

Answer: B

Explanation:

Post-route extraction reveals real RC values. Short data paths become even faster, worsening hold.

Question 19

What is the primary advantage of using H-tree clock topology?

- A) Minimum skew
- B) Minimum power
- C) Minimum latency
- D) Reduced area

Answer: A

Explanation:

H-tree provides symmetric clock distribution, reducing skew.

Question 20

In multi-voltage designs, level shifters are required when:

- A) Driving same voltage domains
- B) Driving from higher voltage to lower voltage only
- C) Driving between different voltage domains
- D) Clock frequency changes

Answer: C

Explanation:

Level shifters are inserted between different voltage domains to avoid device damage and logic failure.

Question 21

Which of the following increases dynamic power the most?

- A) Decreasing voltage
- B) Increasing frequency
- C) Reducing capacitance
- D) Lowering switching activity

Answer: B

Explanation: Dynamic power $\propto f \times C \times V^2$. Increasing frequency directly increases power.

Question 22

During placement optimization, which factor most affects congestion?

- A) Cell height
- B) Netlist connectivity
- C) Clock period
- D) Setup slack

Answer: B

Explanation: Highly connected modules create routing hotspots.

Question 23

If electromigration violation occurs in a clock net, the best immediate fix is:

- A) Reduce clock frequency
- B) Increase metal width
- C) Decrease skew
- D) Add filler cells

Answer: B

Explanation: Increasing metal width reduces current density.

Question 24

What is the major reason for using NDR (Non-Default Rules) in clock routing?

- A) Reduce routing layers
- B) Reduce skew and improve signal integrity
- C) Increase congestion
- D) Reduce cell count

Answer: B

Explanation: NDR uses wider wires and spacing to improve clock signal quality.

Question 25

In advanced FinFET nodes, which variability significantly affects timing closure?

- A) Channel length variation
- B) Random dopant fluctuation
- C) Line Edge Roughness (LER)
- D) Ideal voltage supply

Answer: C

Explanation: LER significantly affects transistor characteristics at small geometries.

Question 26

During CTS, why are hold buffers sometimes inserted intentionally?

- A) To reduce power
- B) To increase clock latency
- C) To fix short data path issues
- D) To reduce IR drop

Answer: C

Explanation: Hold buffers increase data path delay to avoid hold violations.

Question 27

Which parameter is most critical in OCV (On-Chip Variation) modeling?

- A) Clock frequency
- B) Process variation
- C) Routing congestion
- D) Netlist size

Answer: B

Explanation: OCV accounts for process variations affecting delays.

Question 28

What is the primary objective of ECO routing?

- A) Redesign entire layout
- B) Modify small changes without disturbing full routing
- C) Reduce area
- D) Improve power grid

Answer: B

Explanation: ECO routing fixes localized changes while preserving rest of layout.

Question 29

Which technique helps reduce crosstalk delay?

- A) Increase victim wire length
- B) Reduce spacing between nets
- C) Insert shielding or increase spacing
- D) Increase switching activity

Answer: C

Explanation: Spacing or shielding reduces coupling capacitance.

Question 30

In MCMM optimization, why is a common clock tree used across modes?

- A) Reduce cell count
- B) Ensure consistent skew across modes
- C) Increase setup margin
- D) Reduce routing layers

Answer: B

Explanation: A common clock tree maintains consistent skew and simplifies timing closure across modes.



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